

Future-Work Items as Controlled Experiments

Propositions, tests, and verdicts for four extensions of the routing audit

Prepared for review — routing-audit thread

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Abstract

The poster’s future-work section lists five limitations of the current audit: recovery of the full partition rather than the boundary (needs certification); probes that perturb the full input vector rather than location only; $K > 2$ regimes; non-scalar outputs; and non-geographic anchor graphs. This note takes the four that admit controlled study now, states a falsifiable proposition for each *before* running anything, tests it on synthetic models with known ground truth, and records the verdict — including two propositions that were partly wrong, why, and what repairs them. One operator recurs across three of the four studies: *trimmed local-linear residualization* of responses on the probe displacement. Plain OLS residualization destroys the signal it is meant to expose (it absorbs $\approx 64\%$ of the gap at balanced mixing; closed form below); the trimmed variant keeps the fitted plane on the majority branch and (i) makes full-vector probing dimension-independent (E1), (ii) restores validity on clumpy covariate manifolds where the raw dip false-positives at 0.49 (E4), and (iii) even raises power on well-behaved manifolds when the smooth trend is strong ($0.14 \rightarrow 0.90$). All experiments are seeded and reproducible (`sim/ext*.py`); the certification item is discussed but not tested, as it requires the Stage-B machinery that remains open.

1 Setup common to all experiments

Protocol as in the consolidated note: naive scan over 3 scales, Hartigan dip per test with Bonferroni over the tests actually made, per-scale minimum-signal abstention, $m=1000$ queries per test, $\alpha = 0.05$, gate penalty $\Delta = 0.30$, observation noise $\tau = 0.02$, seeds fixed per cell. Every gated model has a matched no-gate control; power is reported on anchors whose probes actually reach a boundary ($\pi_{\text{true}} \geq 0.05$) and false positives on the matched control. “Detectable” and π_{true} follow the corrected (mixing-fraction) ground truth.

2 E1 — Perturbing the full input vector

Proposition P1. Gate on coordinate 1 at distance d ; honest slopes β on every coordinate; dimension D . (a) An isotropic probe with *fixed total budget* (per-coordinate sd $\sigma/\sqrt{D}$) crosses with rate $\Phi(-d\sqrt{D}/\sigma)$: detectability collapses in D . (b) With *fixed per-coordinate* scale σ , crossing is D -independent but within-branch response spread grows as $\sigma\beta\sqrt{D-1}$, masking the gap. (c) OLS residualization of y on the displacement removes the off-axis spread and recovers most of (b)’s loss. (d) Coordinate-targeted scanning restores 1-D power at $D\times$ cost.

Design. $D \in \{2, 4, 8, 16\}$, 100 anchors spanning the boundary band, five probe strategies, gated + control (16,000 audits).

Results.

power	$D=2$	$D=4$	$D=8$	$D=16$	FP	queries
isotropic, fixed total	1.00	0.95	0.83*	0.86*	0.00	3000
isotropic, per-coord	1.00	0.98	0.80	0.55	0.00	3000
+ OLS residualization	0.03	0.05	0.01	0.02	0.00	3000
+ trimmed residualization	0.76	0.75	0.74	0.74	0.00	3000
coordinate scan	1.00	1.00	1.00	1.00	0.00	3000D

*conditional on detectability, which itself collapses: detectable anchors fall $100 \rightarrow 78 \rightarrow 56$ at $D = 8, 16$ — the loss appears as vanishing π , exactly as P1(a) predicts. Empirical median π_{true} matches $\Phi(-d\sqrt{D}/\sigma)$ to three decimals at every D (e.g. $D=16$: 0.063 vs. 0.067).

Verdict. (a), (b), (d): **Confirmed**. (a) is quantitative, not just directional. (c): **Refuted**. — and the failure is instructive. The step is strongly correlated with δ_1 , so the OLS plane absorbs most of it: with $z = \delta_1/\sigma$ and a boundary through the anchor, the residual between-branch separation is

$$\Delta (1 - \varphi(0) (\mathbb{E}[z | z > 0] - \mathbb{E}[z | z < 0])) = \Delta (1 - 2\varphi(0)\sqrt{2/\pi}) \approx 0.36 \Delta,$$

while the tilted fit adds within-branch spread $\approx 0.24 \Delta$: separation ~ 1.5 component-sds, and the dip goes blind. The repair — P1(c'), added after the falsification and then tested — is *trimmed* residualization (fit, drop the top 25% of |residual|, refit, residualize all): the plane stays on the majority branch, the minority jump survives at full size, and power is restored to 0.74–0.76 *independent of D*. At $D=16$ this beats the raw probe (0.55) at equal query cost. The practical ordering at fixed budget: coordinate scan when $D \times$ queries are affordable; trimmed residualization otherwise.

3 E2 — $K > 2$ regimes

Proposition P2. Three regimes, $f(x) = \beta x - 0.25 \mathbf{1}[x \geq \frac{1}{3}] - 0.50 \mathbf{1}[x \geq \frac{2}{3}]$ (levels 0, -0.25 , -0.75). (a) BIC-selected mixtures at the most significant scale recover the number of regimes the probe overlaps; (b) adjacent-gap estimates are consistent where the probe resolves them; (c) in the scalar case, local-to-global label matching reduces to clustering trend-adjusted component levels (a mini-synchronization requiring no assignment machinery), because within-regime drift over a probe width is small against the gaps.

Results. 150 anchors; scales $\{0.02, 0.09, 0.40\}$, the top one spanning both boundaries from mid-domain.

- (a) **Refined**. Raw BIC *over*-partitions: accuracy 0.64, with $k_{\text{true}}=3 \rightarrow \widehat{K}=4$ dominant — trend smear inside each branch is non-Gaussian and BIC pays for it with extra components. Applying the audit's own registered effect-size floor as a *merge rule* (components closer than 0.15 are one regime) lifts accuracy to 0.75 and flips the error direction to mild under-counting — the conservative direction for an audit claim.
- (b) **Confirmed**. at boundary-local scales: median gap 0.228 at the 0.25 boundary, 0.477 at the 0.50 boundary. At the wide scale the three-mode fits recover the total span (0.61 vs. 0.75) but attribute it poorly across the two gaps. Design lesson: *K-counting and gap estimation want different scales* — count wide, estimate narrow.

- (c) **Confirmed.**, with one repair: per-anchor slope estimates absorb the step near boundaries (the E1 mechanism again), so the trend adjustment must use the *global median* slope ($\hat{\beta} = 0.134$, true 0.15). Then clustering the adjusted levels labels components at **0.997** accuracy ($n = 330$ components), despite the cluster centers themselves being biased ($0/-0.19/-0.64$ vs. $0/-0.25/-0.75$): matching is much easier than estimation, which is exactly why the scalar case avoids SYNC1–7.

4 E3 — Non-scalar outputs

Proposition P3. Responses in $\mathbb{R}^V$ ($V=3$), within-branch variation along a direction u (slope $\beta=3$), penalty Δv with $\theta = \angle(u, v)$. (a) The dip on PC1 fails as $\theta \rightarrow 90^\circ$ (PC1 locks to u); (b) top-2 PCs with Bonferroni restore power when variation is rank ≤ 2 ; (c) output axes work only if v is axis-aligned, random projections pay $\sim 1/\sqrt{V}$; (d) responsibilities from the best projection transfer to $\mathbb{R}^V$ and the component-conditional mean difference recovers the penalty *vector*.

Results (power at $V=3$; FP = 0.00 throughout).

θ	0°	30°	60°	90°
PC1 only	0.00	0.00	0.62	1.00
top-2 PCs (Bonf.)	0.00	1.00	0.67	1.00
output axes (Bonf.)	0.00	1.00	1.00	1.00
8 random proj.	0.00	0.97	1.00	1.00

Verdict. Refined. — the geometry is different, and cleaner, than P3(a) assumed. What matters is the decomposition of the penalty into components *parallel* and *orthogonal* to the within-branch variation. The parallel component hides inside the trend (the scalar trend-masking phenomenon); the orthogonal component is nearly noise-free. Hence: at $\theta = 0$ *nothing* detects — a genuine identifiability limit, not a projection failure; at intermediate angles PC1 fails (0.00 at 30°) because it sees only the masked parallel part, while any projection with weight on the orthogonal component succeeds; and at $\theta = 90^\circ$ PC1 *succeeds*, contrary to P3(a), because the scale ladder immunizes it: within-branch variance shrinks like $(\beta\sigma)^2$ as σ decreases while the gap stays Δ , so at the smallest scale the between-branch direction dominates and PC1 rotates onto v . P3(b) holds with a caveat (at 60° PC2 catches only $\Delta \sin \theta$); axes and random projections were the most robust overall here. (d) **Confirmed.** after one correction with the same flavor as E1’s: the raw group-mean difference is contaminated by the trend times the difference in where the groups sit ($\|\hat{\Delta}\| = 0.57$, cosine 0.52); defining the estimand as the difference of *group-conditional local-linear predictions at the anchor* recovers $\|\hat{\Delta}\| = 0.297\text{--}0.301$ with cosine 1.000 to v , at $V=3$ and $V=8$. The open item is estimated (non-isotropic) whitening.

5 E4 — Non-geographic anchor graphs (clumpy covariates)

Proposition P4. Covariates drawn from a 6-cluster mixture (the attribute-space analogue of construction booms and standard floor counts), strong smooth attribute trend, A11 satisfied. (a) On-manifold probes at scales comparable to the inter-cluster spacing inherit the data’s multimodality: the honest model is flagged far above nominal (the A12 violation, reproduced in vitro). (b) Local-linear residualization restores near-nominal validity while the jump retains power. (c) On a uniform

manifold the raw dip is already valid: the failure belongs to the data distribution, not to on-manifold probing.

Results (130 anchors per cell; power on detectable anchors).

manifold	method	FP (honest)	power
clumpy	raw	0.49	0.86
clumpy	OLS residualization	0.00	0.16
clumpy	trimmed residualization	0.00	0.57
uniform	raw	0.00	0.14 [†]
uniform	trimmed residualization	0.00	0.90

[†]the strong attribute trend masks the gap at large scales even on a uniform manifold — the scalar trend-masking limit, not an A12 effect.

Verdict. (a), (c): **Confirmed.** — FP 0.49 vs. 0.00 on the same model with the same probes, differing only in the data distribution. (b): **Refined.** — plain OLS residualization does fix validity but pays the E1 gap-absorption cost (power 0.16); the trimmed variant keeps validity at FP 0.00 *and* power 0.57. The unexpected result is the uniform row: trimmed residualization raises power $0.14 \rightarrow 0.90$ by removing the trend that was masking the gap. This converts the “trend width vs. gap” power limit into a “residual width vs. gap” limit — a strictly better trade wherever the honest surface is locally linear (A11).

6 Cross-cutting conclusions

1. **Trimmed local-linear residualization should likely become a standard pipeline stage.** It appears independently as the repair in E1 (dimension-independence), E4 (A12 validity), and E4-uniform (trend unmasking), and its failure mode is now understood in closed form (plain OLS absorbs $1 - 2\varphi(0)\sqrt{2/\pi} \approx 64\%$ of the gap at balanced mixing). Open: behaviour under curved honest surfaces near the A11 margin, and at mixing fractions near 0.5 where trimming is ambiguous.
2. **The registered effect-size floor does double duty** as a component-merge rule for K selection, shifting BIC’s over-partitioning to conservative under-counting (E2).
3. **Scale ladders have two distinct jobs** — count regimes wide, estimate gaps narrow (E2); and the small end of the ladder is what immunizes PC1 against alignment failure (E3).
4. **Two genuine identifiability limits surfaced:** a penalty parallel to strong within-branch variation is invisible to any projection at any angle (E3, $\theta = 0$); and fixed-total-budget probing in high dimension destroys detectability itself, not just power (E1(a), with the crossing-rate formula verified to three decimals).
5. **Recovery estimands must be group-conditional predictions at the anchor** (E3(d), and implicitly the scalar $\hat{\Delta}$): raw component-mean differences mix the penalty with the trend across the two groups’ supports.

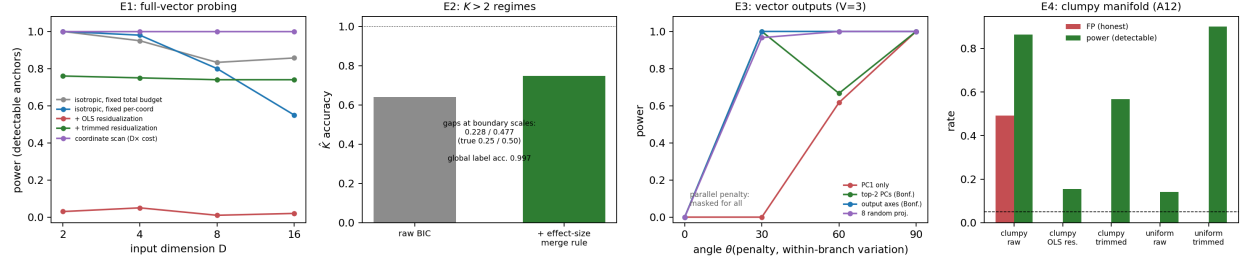


Figure 1: E1–E4 headline results. E1: power vs. dimension by probe strategy; E2: $\widehat{K}$ accuracy before/after the merge rule; E3: power vs. penalty angle by projection strategy; E4: honest-model FP and gated-model power on clumpy vs. uniform manifolds.

7 Not tested here

Full partition via certification (poster item 1) is not a simulation-sized question: it needs bridge-probe certification (CERT1–4), the abstention bookkeeping, and ideally the general- K synchronization layer, i.e. the Stage-B machinery that the consolidated note lists as the largest open item. The E2 result narrows it slightly: in the scalar case, global labels come from level clustering at 0.997 accuracy, so the certification problem reduces to marking *territory* (where between-anchor edges are supported by probes), not to label agreement.

8 Reproducibility

Code: `sim/ext_core.py`, `sim/ext1_fullvec.py` ... `sim/ext4_nongeo.py`, driver `sim/run_ext.py`; seeded, resumable, per-cell CSVs; summaries via `summarize`. Anchor-level outputs: `e1_anchors.csv` ... `e4_anchors.csv`; summaries: `e1_summary.csv`, `e2_confusion*.csv`, `e2_levels.csv`, `e3_summary.csv`, `e4_summary.csv`. Dependencies: numpy, pandas, matplotlib (diptest optional; the Monte-Carlo dip fallback is seeded).